

Topology MATM36: Course Summary of Important Results

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Metric Spaces

Definition 1. Let X be a topological space and $A \subseteq X$. A point $x \in X$ is called an adherent point of A if every open neighbourhood U of x satisfies

$$U \cap A \neq \emptyset.$$

In the special case where X is a metric space with metric d , this is equivalent to requiring that for every $r > 0$,

$$B_r(x) \cap A \neq \emptyset.$$

Definition 2. Let X be a topological space and $A \subseteq X$. The set A is called dense in X if every nonempty open set $U \subseteq X$ satisfies

$$U \cap A \neq \emptyset.$$

Equivalently, A is dense in X if $\overline{A} = X$.

Definition 3. Let X be a topological space and $A \subseteq X$. The set A is called nowhere dense in X if

$$\text{int}(\overline{A}) = \emptyset.$$

Definition 4. A topological space X is separable if: $\exists D \subseteq X : D$ countable and dense.

Baire Category Theorem

Theorem 1 (Baire Category Theorem). Let (X, d) be a complete metric space and let $\{U_n\}_{n=1}^\infty$ be a sequence of dense open subsets of X . Then $\bigcap_{n=1}^\infty U_n$ is dense in X .

Proof. Fix a nonempty open ball $B(x, \varepsilon) \subset X$.

Since U_1 is dense and open, choose $y_1 \in U_1 \cap B(x, \varepsilon)$ and $r_1 > 0$ such that

$$\overline{B}(y_1, r_1) \subset U_1 \cap B(x, \varepsilon), \quad r_1 < 1.$$

Inductively, having chosen y_n, r_n , use that U_{n+1} is dense and open to pick $y_{n+1} \in U_{n+1} \cap B(y_n, r_n)$ and $r_{n+1} > 0$ such that

$$\overline{B}(y_{n+1}, r_{n+1}) \subset U_{n+1} \cap B(y_n, r_n), \quad r_{n+1} < \frac{1}{n+1}.$$

Then the closed balls are nested:

$$\overline{B}(y_{n+1}, r_{n+1}) \subset \overline{B}(y_n, r_n) \subset \cdots \subset \overline{B}(y_1, r_1) \subset B(x, \varepsilon),$$

and for $m > n$ we have $y_m \in B(y_n, r_n)$, hence $d(y_m, y_n) < r_n \rightarrow 0$. Thus (y_n) is Cauchy, so by completeness $y_n \rightarrow y \in X$.

For each fixed n , all tails of the sequence lie in $\overline{B}(y_n, r_n)$, so the limit satisfies $y \in \overline{B}(y_n, r_n) \subset U_n$. Hence $y \in \bigcap_{n=1}^{\infty} U_n$, and also $y \in \overline{B}(y_1, r_1) \subset B(x, \varepsilon)$. Therefore

$$B(x, \varepsilon) \cap \bigcap_{n=1}^{\infty} U_n \neq \emptyset,$$

so the intersection is dense. □

Remark 1. From the definitions it follows that a set $B \subseteq X$ satisfies $\text{int}(B) = \emptyset \iff X \setminus B$ is dense in X . In particular, if A is nowhere dense, then $X \setminus \overline{A}$ is open and dense in X .

Corollary 1. Let (X, d) be a complete metric space and let $\{F_n\}_{n=1}^{\infty}$ be a sequence of nowhere dense subsets of X . Then

$$\text{int}\left(\bigcup_{n=1}^{\infty} F_n\right) = \emptyset.$$

Proof. Since F_n is nowhere dense, $U_n := X \setminus \overline{F_n}$ is open and dense; by Baire, $\bigcap_{n=1}^{\infty} U_n = \bigcup_{n=1}^{\infty} \overline{F_n}$ is dense, hence nonempty in every nonempty open set. □

Finite Products of Metric Spaces

Definition 5. Let (X_k, d_k) , $k = 1, \dots, n$, be metric spaces and set

$$X = X_1 \times \cdots \times X_n.$$

A metric d on X is said to be a product metric if convergence in (X, d) is coordinatewise,

$$x^{(m)} = (x_1^{(m)}, \dots, x_n^{(m)}) \rightarrow x = (x_1, \dots, x_n)$$

in X if and only if for each $k = 1, \dots, n$,

$$x_k^{(m)} \rightarrow x_k \quad \text{in } X_k.$$

Theorem 2. Suppose d is a metric on $X = X_1 \times \cdots \times X_n$ satisfying the above coordinatewise convergence property. Then the open sets in (X, d) are precisely the unions of product sets

$$U_1 \times \cdots \times U_n,$$

where each U_k is open in X_k .

Proof. First, if $E_k \subset X_k$ is closed for each k , then

$$E_1 \times \cdots \times E_n$$

is closed in X . Indeed, if $x^{(m)} \in E_1 \times \cdots \times E_n$ and $x^{(m)} \rightarrow x$ in X , then by coordinatewise convergence, $x_k^{(m)} \rightarrow x_k$ in X_k . Since each E_k is closed, $x_k \in E_k$, hence $x \in E_1 \times \cdots \times E_n$.

Now let $U_k \subset X_k$ be open. Then

$$V^{(k)} := X_1 \times \cdots \times X_{k-1} \times U_k \times X_{k+1} \times \cdots \times X_n$$

is open in X , since its complement is of the form

$$X_1 \times \cdots \times (X_k \setminus U_k) \times \cdots \times X_n,$$

which is closed by the previous remark. Hence

$$U_1 \times \cdots \times U_n = \bigcap_{k=1}^n V^{(k)}$$

is open in X , and any union of such products is open.

Conversely, let $U \subset X$ be open and $x = (x_1, \dots, x_n) \in U$. We show there exist open sets $U_k \subset X_k$ such that

$$x \in U_1 \times \cdots \times U_n \subset U.$$

Suppose not. Then for each $m \in \mathbb{N}$,

$$B_{X_1}(x_1, 1/m) \times \cdots \times B_{X_n}(x_n, 1/m) \not\subset U.$$

Thus there exists $x^{(m)} \notin U$ with

$$d_k(x_k^{(m)}, x_k) < \frac{1}{m}, \quad k = 1, \dots, n.$$

Then $x_k^{(m)} \rightarrow x_k$ for each k , hence $x^{(m)} \rightarrow x$ in X . Since $X \setminus U$ is closed and each $x^{(m)} \in X \setminus U$, we conclude $x \in X \setminus U$, a contradiction. Therefore such open sets U_k exist, and U is a union of product sets. $\square$

Theorem 3 (5.1). *For a metric space X , the following are equivalent:*

- (i) X is compact;
- (ii) every sequence in X has a convergent subsequence;
- (iii) X is complete and totally bounded.

Proof outline with key implications. We prove the cycle of implications

$$(i) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow (ii) \Rightarrow (i).$$

Implication (i) $\Rightarrow$ (ii). Assume X is compact. Let (y_j) be a sequence in X . We show that (y_j) has a convergent subsequence.

First claim that there exists $x \in X$ such that every ball $B(x, \varepsilon)$ contains infinitely many terms of the sequence. Suppose not. Then for each $x \in X$ there exists $\varepsilon(x) > 0$ such that $B(x, \varepsilon(x))$ contains only finitely many terms of (y_j) . The family

$$\{B(x, \varepsilon(x)) : x \in X\}$$

is an open cover of X . By compactness, there is a finite subcover

$$X = B(x_1, \varepsilon(x_1)) \cup \cdots \cup B(x_m, \varepsilon(x_m)).$$

But each ball contains only finitely many terms of the sequence, so altogether only finitely many terms of (y_j) lie in X , which is impossible. Hence such an x exists.

Now choose inductively indices

$$j_1 < j_2 < j_3 < \cdots$$

such that

$$y_{j_n} \in B(x, 1/n).$$

Then $y_{j_n} \rightarrow x$, so (y_j) has a convergent subsequence.

Comment. The key idea is that compactness prevents the sequence from avoiding every point at every scale.

Implication $(ii) \Rightarrow (iii)$. Assume every sequence in X has a convergent subsequence. We prove first that X is complete and then that it is totally bounded.

Completeness. Let (x_n) be a Cauchy sequence in X . By (ii), it has a convergent subsequence

$$x_{n_k} \rightarrow x.$$

A standard lemma says that if a Cauchy sequence has a convergent subsequence, then the whole sequence converges to the same limit. Hence $(x_n) \rightarrow x$, and X is complete.

Total boundedness. Fix $\varepsilon > 0$. We show that X can be covered by finitely many open balls of radius ε .

Suppose not. Choose $y_1 \in X$. Since $B(y_1, \varepsilon) \neq X$, choose

$$y_2 \in X \setminus B(y_1, \varepsilon).$$

Having chosen $y_1, \dots, y_n$, if

$$X \neq \bigcup_{j=1}^n B(y_j, \varepsilon),$$

choose

$$y_{n+1} \in X \setminus \bigcup_{j=1}^n B(y_j, \varepsilon).$$

Then

$$d(y_i, y_j) \geq \varepsilon \quad (i \neq j).$$

So the sequence (y_j) has no Cauchy subsequence, hence no convergent subsequence, contradicting (ii). Therefore the process must terminate after finitely many steps, and X is covered by finitely many balls of radius ε . Thus X is totally bounded.

Comment. This is one of the most important implications: failure of total boundedness lets one construct an ε -separated sequence, and such a sequence can never have a convergent subsequence.

Implication $(iii) \Rightarrow (ii)$. Assume X is complete and totally bounded. Let (x_j) be a sequence in X .

We first show that (x_j) has a Cauchy subsequence.

For each $k \in \mathbb{N}$, since X is totally bounded, there exist finitely many balls of radius $1/k$ that cover X . In particular, there exists a finite collection of balls

$$B(x_{k,1}, 1/k), \dots, B(x_{k,N_k}, 1/k)$$

such that

$$X = \bigcup_{i=1}^{N_k} B(x_{k,i}, 1/k).$$

We now construct a sequence of nested subsequences. Since the sequence (x_j) has infinitely many terms and only finitely many balls of radius $1/k$, there exists a ball B_1 of radius $1/k$ that contains infinitely many terms of (x_j) . Let $(x_j^{(1)})$ be the subsequence consisting of all such terms.

Next, consider the covering by balls of radius $1/2$. Since $(x_j^{(1)})$ is still infinite, there exists a ball B_2 of radius $1/2$ containing infinitely many terms of $(x_j^{(1)})$. Let $(x_j^{(2)})$ be this subsequence.

Proceeding inductively, for each k we obtain a subsequence $(x_j^{(k)})$ of $(x_j^{(k-1)})$ such that all its terms lie in some ball B_k of radius $1/k$.

Thus we obtain nested subsequences

$$(x_j) \supset (x_j^{(1)}) \supset (x_j^{(2)}) \supset \dots$$

Now define a new sequence (y_n) by taking the diagonal subsequence:

$$y_n := x_n^{(n)}.$$

We claim that (y_n) is Cauchy. Let $\varepsilon > 0$. Choose $k \in \mathbb{N}$ such that $1/k < \varepsilon$. By construction, all terms of $(x_j^{(k)})$ lie in a ball of radius $1/k$. For $n \geq k$, we have $y_n \in (x_j^{(k)})$, so y_n lies in this ball. Hence for all $n, m \geq k$,

$$d(y_n, y_m) < \frac{2}{k} < 2\varepsilon.$$

Thus (y_n) is Cauchy.

Since X is complete, (y_n) converges in X . Therefore (x_j) has a convergent subsequence.

Implication $(ii) \Rightarrow (i)$. Assume every sequence in X has a convergent subsequence. Since by $(ii) \Rightarrow (iii)$ the space is totally bounded, it is separable; and every separable metric space is second-countable. Hence every open cover has a countable subcover. It therefore suffices to prove that every countable open cover has a finite subcover.

Let

$$X = \bigcup_{n=1}^{\infty} U_n$$

be a countable open cover. Suppose no finite subfamily covers X . Then for each m choose

$$x_m \in X \setminus (U_1 \cup \dots \cup U_m).$$

By (ii), the sequence (x_m) has a convergent subsequence

$$x_{m_k} \rightarrow x \in X.$$

Since the U_n cover X , there exists r such that $x \in U_r$. As U_r is open and $x_{m_k} \rightarrow x$, we have $x_{m_k} \in U_r$ for all sufficiently large k . But whenever $m_k \geq r$, the choice of x_{m_k} gives

$$x_{m_k} \notin U_1 \cup \cdots \cup U_{m_k},$$

hence in particular $x_{m_k} \notin U_r$, a contradiction. Therefore some finite subfamily already covers X , so X is compact.

Comment. The contradiction comes from choosing points left out by larger and larger finite parts of the cover, then using sequential compactness to force a limit point into one of the open sets. □

Theorem 4 (Heine–Borel Theorem). *For a subspace $E \subseteq \mathbb{R}^n$, the following are equivalent:*

- (i) E is compact;
- (ii) every sequence in E has a convergent subsequence with limit in E ;
- (iii) E is closed and bounded.

Proof. By the general compactness theorem for metric spaces, for any metric space the following are equivalent: compactness, sequential compactness, and completeness together with total boundedness.

Now let $E \subseteq \mathbb{R}^n$.

If E is closed and bounded, then E is complete (since a subspace of $\mathbb{R}^n$ is complete if and only if it is closed) and totally bounded (since a subset of $\mathbb{R}^n$ is totally bounded if and only if it is bounded). Hence E is compact, and therefore every sequence in E has a convergent subsequence.

Conversely, if E is compact, then by the same general theorem E is complete and totally bounded. Since E is a subspace of $\mathbb{R}^n$, completeness implies that E is closed, and total boundedness implies that E is bounded. Thus E is closed and bounded.

Therefore (i), (ii), and (iii) are equivalent. □

Theorem 5 (Uniform Continuity Theorem). *Let (X, d) and (Y, ρ) be metric spaces, and suppose that X is compact. Then every continuous function $f : X \rightarrow Y$ is uniformly continuous.*

Proof. Suppose f is not uniformly continuous. Then there exists $\varepsilon > 0$ such that for every $m \in \mathbb{N}$ there are points $x_m, z_m \in X$ with

$$d(x_m, z_m) < \frac{1}{m} \quad \text{but} \quad \rho(f(x_m), f(z_m)) \geq \varepsilon.$$

Since X is compact, the sequence (x_m) has a convergent subsequence; passing to that subsequence, we may assume

$$x_m \rightarrow x \in X.$$

Because $d(x_m, z_m) \rightarrow 0$, also

$$z_m \rightarrow x.$$

By continuity of f at x ,

$$f(x_m) \rightarrow f(x) \quad \text{and} \quad f(z_m) \rightarrow f(x).$$

Hence

$$\rho(f(x_m), f(z_m)) \leq \rho(f(x_m), f(x)) + \rho(f(x), f(z_m)) \rightarrow 0,$$

which contradicts

$$\rho(f(x_m), f(z_m)) \geq \varepsilon.$$

Therefore f is uniformly continuous. □

Theorem 6 (Theorem 7.5). *Let X and Y be normed spaces and let $T : X \rightarrow Y$ be linear. The following are equivalent:*

- (i) T is continuous;
- (ii) T is continuous at 0;
- (iii) T is bounded, i.e. $\exists C > 0 : \|T(x)\| \leq C\|x\|$ for all $x \in X$.

Proof. (i) $\Rightarrow$ (ii) is immediate.

(ii) $\Rightarrow$ (iii): Since T is continuous at 0, there exists $\delta > 0$ such that

$$\|x\| \leq \delta \Rightarrow \|T(x)\| \leq 1.$$

For arbitrary $x \neq 0$, set $y = \frac{\delta}{\|x\|}x$, so $\|y\| = \delta$ and hence

$$\|T(y)\| \leq 1.$$

By linearity,

$$\|T(x)\| = \frac{\|x\|}{\delta} \|T(y)\| \leq \frac{1}{\delta} \|x\|.$$

Thus T is bounded.

(iii) $\Rightarrow$ (i): If $\|T(x)\| \leq C\|x\|$, then for all $x, x_0 \in X$,

$$\|T(x) - T(x_0)\| = \|T(x - x_0)\| \leq C\|x - x_0\|,$$

so T is Lipschitz, hence continuous. □

Theorem 7 (Principle of Uniform Boundedness). *Let X and Y be Banach spaces and let $\{T_\alpha\}_{\alpha \in A}$ be a family of continuous linear operators $T_\alpha : X \rightarrow Y$. Suppose that for each $x \in X$,*

$$\sup_{\alpha \in A} \|T_\alpha x\| < \infty.$$

Then

$$\sup_{\alpha \in A} \|T_\alpha\| < \infty.$$

Proof. For $n \in \mathbb{N}$, define

$$E_n := \left\{ x \in X : \sup_{\alpha \in A} \|T_\alpha x\| \leq n \right\}.$$

Then each E_n is closed, $E_n \subseteq E_{n+1}$, and by assumption

$$X = \bigcup_{n=1}^{\infty} E_n.$$

By the Baire Category Theorem, some E_m has nonempty interior. Thus there exist $x_0 \in X$ and $\varepsilon > 0$ such that

$$B(x_0, \varepsilon) \subseteq E_m.$$

Let $c := \sup_{\alpha \in A} \|T_\alpha x_0\| < \infty$. If $\|x\| < 1$, then $x_0 + \varepsilon x \in E_m$, hence

$$\|T_\alpha(x_0 + \varepsilon x)\| \leq m.$$

Thus

$$\varepsilon \|T_\alpha x\| = \|T_\alpha(x_0 + \varepsilon x) - T_\alpha(x_0)\| \leq m + c.$$

Therefore

$$\|T_\alpha x\| \leq \frac{m+c}{\varepsilon} \quad \text{for all } \|x\| < 1.$$

Hence $\|T_\alpha\| \leq \frac{m+c}{\varepsilon}$ for all $\alpha \in A$, and so

$$\sup_{\alpha \in A} \|T_\alpha\| < \infty.$$

□

Topologies

Definition 6. Let X be a set. A topology on X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ of subsets of X such that

(i) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$,

(ii) if $\{U_\alpha\}_{\alpha \in I} \subseteq \mathcal{T}$, then

$$\bigcup_{\alpha \in I} U_\alpha \in \mathcal{T},$$

(iii) if $U_1, \dots, U_n \in \mathcal{T}$, then

$$\bigcap_{i=1}^n U_i \in \mathcal{T}.$$

The pair $(X, \mathcal{T})$ is called a topological space. The sets $U \in \mathcal{T}$ are called open sets. A subset $F \subseteq X$ is called closed if its complement is open.

Separation Axioms

Definition 7 (T_0 space). A topological space $(X, \mathcal{T})$ is called a T_0 -space if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that

$$x \in U, y \notin U \quad \text{or} \quad y \in U, x \notin U.$$

Definition 8 (T_1 space). A topological space $(X, \mathcal{T})$ is called a T_1 -space if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that

$$y \in U \quad \text{and} \quad x \notin U.$$

Equivalently, $(X, \mathcal{T})$ is T_1 if and only if every singleton set $\{x\}$ is closed in X .

Definition 9 (T_2 space / Hausdorff). A topological space $(X, \mathcal{T})$ is called a T_2 -space, or Hausdorff, if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \in \mathcal{T}$ such that

$$x \in U, \quad y \in V, \quad U \cap V = \emptyset.$$

Definition 10 (Regular space). A topological space $(X, \mathcal{T})$ is called regular if for every closed set $F \subseteq X$ and every point $x \in X \setminus F$, there exist open sets $U, V \in \mathcal{T}$ such that

$$x \in U, \quad F \subseteq V, \quad U \cap V = \emptyset.$$

Definition 11 (T_3 space). A topological space $(X, \mathcal{T})$ is called a T_3 -space if it is both T_1 and regular.

Definition 12 (Normal space). A topological space $(X, \mathcal{T})$ is called normal if for any two disjoint closed sets $F, G \subseteq X$, there exist open sets $U, V \in \mathcal{T}$ such that

$$F \subseteq U, \quad G \subseteq V, \quad U \cap V = \emptyset.$$

Definition 13 (T_4 space). A topological space $(X, \mathcal{T})$ is called a T_4 -space if it is both T_1 and normal.

Theorem 8. Every compact Hausdorff space is normal. In particular, every compact Hausdorff space is a T_4 -space.

Proof. Let $F, G \subseteq X$ be disjoint closed sets. Since X is Hausdorff, for each $x \in F$ and $y \in G$ there exist disjoint open sets $U_{x,y}$ and $V_{x,y}$ such that

$$x \in U_{x,y}, \quad y \in V_{x,y}, \quad U_{x,y} \cap V_{x,y} = \emptyset.$$

For fixed $x \in F$, the sets $\{V_{x,y}\}_{y \in G}$ form an open cover of G . Since G is closed in the compact space X , it is compact, so we may choose a finite subcover

$$G \subseteq V_{x,y_1} \cup \cdots \cup V_{x,y_n}.$$

Let

$$U_x := U_{x,y_1} \cap \cdots \cap U_{x,y_n}, \quad V_x := V_{x,y_1} \cup \cdots \cup V_{x,y_n}.$$

Then $x \in U_x$, $G \subseteq V_x$, and $U_x \cap V_x = \emptyset$.

The family $\{U_x\}_{x \in F}$ is an open cover of F . Since F is compact, choose a finite subcover

$$F \subseteq U_{x_1} \cup \cdots \cup U_{x_m}.$$

Define

$$U := U_{x_1} \cup \cdots \cup U_{x_m}, \quad V := V_{x_1} \cap \cdots \cap V_{x_m}.$$

Then U and V are open, $F \subseteq U$, $G \subseteq V$, and $U \cap V = \emptyset$. Hence X is normal. $\square$

Remark 2 (Continuity criterion in metric spaces). Let $(X, \mathcal{T})$ be a topological space and (Y, d) a metric space. A function $f : X \rightarrow Y$ is continuous at $x \in X$ if and only if for every $\varepsilon > 0$ there exists a neighbourhood U of x such that

$$d(f(y), f(x)) < \varepsilon \quad \text{for all } y \in U.$$

Equivalently, $f(U) \subseteq B(f(x), \varepsilon)$.

Theorem 9 (Uniform limit preserves continuity). *Let $(X, \mathcal{T})$ be a topological space and (Y, d) a metric space. Suppose (f_n) is a sequence of functions $f_n : X \rightarrow Y$ converging uniformly to $f : X \rightarrow Y$. If each f_n is continuous at $x \in X$, then f is continuous at x .*

Proof. We first recall the following characterization of continuity at a point: f is continuous at x if and only if for every $\varepsilon > 0$ there exists a neighbourhood U of x such that

$$d(f(y), f(x)) < \varepsilon \quad \text{for all } y \in U.$$

Now let $\varepsilon > 0$. By uniform convergence, there exists $N \in \mathbb{N}$ such that

$$d(f_N(y), f(y)) < \frac{\varepsilon}{3} \quad \text{for all } y \in X.$$

Since f_N is continuous at x , there exists a neighbourhood U of x such that

$$d(f_N(y), f_N(x)) < \frac{\varepsilon}{3} \quad \text{for all } y \in U.$$

Then for all $y \in U$,

$$d(f(y), f(x)) \leq d(f(y), f_N(y)) + d(f_N(y), f_N(x)) + d(f_N(x), f(x)) < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

By the above characterization, f is continuous at x . □

Theorem 10. *Every metric space (X, d) is a T_4 -space.*

Proof. First, for each $x \in X$,

$$\{x\} = \bigcap_{r>0} \overline{B(x, r)},$$

so singletons are closed and X is T_1 .

Let $E, F \subseteq X$ be disjoint closed sets. For each $x \in E$, there exists $r(x) > 0$ such that

$$B(x, r(x)) \cap F = \emptyset.$$

Similarly, for each $y \in F$, there exists $r(y) > 0$ such that

$$B(y, r(y)) \cap E = \emptyset.$$

Define

$$U := \bigcup_{x \in E} B\left(x, \frac{r(x)}{2}\right), \quad V := \bigcup_{y \in F} B\left(y, \frac{r(y)}{2}\right).$$

Then U, V are open, $E \subseteq U$, $F \subseteq V$.

Suppose $z \in U \cap V$. Then there exist $x \in E$, $y \in F$ such that

$$d(x, z) < \frac{r(x)}{2}, \quad d(y, z) < \frac{r(y)}{2}.$$

Hence

$$d(x, y) \leq d(x, z) + d(z, y) < \frac{r(x) + r(y)}{2} \leq \max\{r(x), r(y)\}.$$

Thus either $d(x, y) < r(x)$ or $d(x, y) < r(y)$, contradicting the choice of $r(x)$ or $r(y)$. Therefore $U \cap V = \emptyset$, and X is normal. □

Theorem 11 (Urysohn's Lemma). *Let E and F be disjoint closed subsets of a normal space X . Then there exists a continuous function*

$$f : X \rightarrow [0, 1]$$

such that

$$f|_E = 0, \quad f|_F = 1.$$

Proof. Set $V := X \setminus F$, which is open and contains E . Using normality repeatedly, construct for each dyadic rational $r \in (0, 1)$ an open set U_r such that

$$E \subseteq U_r \subseteq V$$

and

$$\overline{U_r} \subseteq U_s \quad \text{whenever } 0 < r < s < 1.$$

Define $f : X \rightarrow [0, 1]$ by

$$f(x) = \begin{cases} 0, & x \in \bigcap_{0 < r < 1} U_r, \\ \sup\{r \in (0, 1) \cap D : x \notin U_r\}, & \text{otherwise,} \end{cases}$$

where D denotes the set of dyadic rationals in $(0, 1)$.

Then $0 \leq f \leq 1$, and since $E \subseteq U_r$ for every r , we have $f = 0$ on E . Also $U_r \subseteq X \setminus F$ for every r , so $f = 1$ on F .

To prove continuity, let $x \in X$ and $\varepsilon > 0$. Choose dyadic rationals r, s such that

$$f(x) - \varepsilon < r < f(x) < s < f(x) + \varepsilon.$$

Then $x \notin \overline{U_r}$ and $x \in U_s$. Hence

$$W := U_s \setminus \overline{U_r}$$

is an open neighbourhood of x . For every $y \in W$ we have $y \notin U_r$, so $f(y) \geq r$, and $y \in U_s$, so $f(y) \leq s$. Thus

$$r \leq f(y) \leq s,$$

and therefore

$$|f(y) - f(x)| < \varepsilon.$$

So f is continuous. □

Theorem 12 (Tietze Extension Theorem). *Let X be a normal topological space, $Y \subseteq X$ closed, and let $f : Y \rightarrow \mathbb{R}$ be bounded and continuous. Then there exists a bounded continuous function $h : X \rightarrow \mathbb{R}$ such that $h|_Y = f$.*

Proof. Assume $|f| \leq c_0$ on Y . Using Urysohn's Lemma, construct a continuous function $g_0 : X \rightarrow \mathbb{R}$ such that

$$|g_0| \leq \frac{c_0}{3} \quad \text{on } X, \quad |f - g_0| \leq \frac{2c_0}{3} \quad \text{on } Y.$$

Inductively, having defined $g_0, \dots, g_{n-1}$, set

$$c_n := \sup_{y \in Y} |f(y) - g_0(y) - \dots - g_{n-1}(y)|.$$

Again by Urysohn, construct g_n such that

$$|g_n| \leq \frac{c_n}{3} \quad \text{on } X, \quad |f - g_0 - \cdots - g_n| \leq \frac{2c_n}{3} \quad \text{on } Y.$$

Then $c_n \leq (2/3)^n c_0$ and $|g_n| \leq (2/3)^n c_0/3$.

The series $\sum g_n$ converges uniformly on X to a continuous function

$$h := \sum_{n=0}^{\infty} g_n.$$

Moreover, on Y ,

$$f = \lim_{n \rightarrow \infty} (g_0 + \cdots + g_n) = h,$$

and h is bounded. Thus h is the desired extension. $\square$

Basis & Subbasis

Definition 14. Let X be a set. A collection $\mathcal{B}$ of subsets of X is called a *basis* for a topology on X if:

- (i) For every $x \in X$ there exists $B \in \mathcal{B}$ such that $x \in B$.
- (ii) If $x \in B_1 \cap B_2$ for $B_1, B_2 \in \mathcal{B}$, then there exists $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

Theorem 13 (Topology generated by a basis). If $\mathcal{B}$ satisfies the two conditions above, then there exists a unique topology $\mathcal{T}$ on X whose open sets are precisely the unions of elements of $\mathcal{B}$. That is,

$$\mathcal{T} = \left\{ \bigcup_{\alpha \in I} B_\alpha : B_\alpha \in \mathcal{B}, I \text{ arbitrary} \right\}.$$

Definition 15. Let X be a set. A collection $\mathcal{S}$ of subsets of X is called a *subbasis* for a topology on X if the union of all sets in $\mathcal{S}$ equals X ,

$$X = \bigcup_{S \in \mathcal{S}} S.$$

The topology generated by $\mathcal{S}$ is obtained by first forming the basis consisting of all finite intersections of elements of $\mathcal{S}$,

$$\mathcal{B} = \{S_1 \cap \cdots \cap S_n : S_i \in \mathcal{S}, n \in \mathbb{N}\},$$

and then taking arbitrary unions of these sets.

The topology generated by the subbasis $\mathcal{S}$ consists of all arbitrary unions of elements of this basis. That is, a set $U \subseteq X$ is open if and only if it can be written as

$$U = \bigcup_{\alpha \in I} \left(\bigcap_{i=1}^{n_\alpha} S_{\alpha,i} \right), \quad S_{\alpha,i} \in \mathcal{S}, \quad n_\alpha \in \mathbb{N}.$$

Theorem 14 (Basis criterion). *A collection $\mathcal{B}$ of subsets of X is a basis for a topology on X if and only if:*

- (i) *For every $x \in X$, there exists $B \in \mathcal{B}$ such that $x \in B$;*
- (ii) *If $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$, then there exists $B_3 \in \mathcal{B}$ such that*

$$x \in B_3 \subseteq B_1 \cap B_2.$$

Proof. ($\Rightarrow$) If $\mathcal{B}$ is a basis, then (i) holds since X is open, and (ii) holds since $B_1 \cap B_2$ is open and can be written as a union of basis elements.

($\Leftarrow$) Suppose (i) and (ii) hold. Let $\mathcal{T}$ be the collection of all unions of sets in $\mathcal{B}$. Then $\emptyset, X \in \mathcal{T}$ and arbitrary unions are in $\mathcal{T}$.

To show closure under finite intersections, let $U, V \in \mathcal{T}$ and $x \in U \cap V$. Then there exist $B_1, B_2 \in \mathcal{B}$ such that

$$x \in B_1 \subseteq U, \quad x \in B_2 \subseteq V.$$

By (ii), there exists $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2 \subseteq U \cap V.$$

Hence

$$U \cap V = \bigcup_{x \in U \cap V} B_3(x),$$

so $U \cap V \in \mathcal{T}$. Thus $\mathcal{T}$ is a topology and $\mathcal{B}$ is a basis. □

Theorem 15 (Lindelöf's Theorem). *Let X be a second-countable topological space. Then every open cover of X admits a countable subcover.*

Proof. Let $\mathcal{B} = \{B_n\}_{n \in \mathbb{N}}$ be a countable basis for X , and let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be an open cover of X .

For each $n \in \mathbb{N}$, since $B_n \subseteq X = \bigcup_{\alpha \in A} U_\alpha$, we may choose $\alpha(n) \in A$ such that

$$B_n \subseteq U_{\alpha(n)}.$$

Then the collection $\{U_{\alpha(n)}\}_{n \in \mathbb{N}}$ is countable.

To see that it covers X , let $x \in X$. Since $\mathcal{B}$ is a basis, there exists n such that $x \in B_n$. By construction, $B_n \subseteq U_{\alpha(n)}$, so $x \in U_{\alpha(n)}$. Hence

$$X = \bigcup_{n=1}^{\infty} U_{\alpha(n)}.$$

Thus $\{U_{\alpha(n)}\}_{n \in \mathbb{N}}$ is a countable subcover. □

Compactness

Theorem 16. *Let S be a compact subset of a Hausdorff space X . For each $x \in X \setminus S$, there exist disjoint open sets $U, V \subseteq X$ such that*

$$x \in U, \quad S \subseteq V.$$

Proof. For each $y \in S$, since X is Hausdorff, there exist disjoint open sets U_y and V_y such that

$$x \in U_y, \quad y \in V_y.$$

Then $\{V_y\}_{y \in S}$ is an open cover of S . By compactness, there exist $y_1, \dots, y_n \in S$ such that

$$S \subseteq V_{y_1} \cup \dots \cup V_{y_n} =: V.$$

Let

$$U := U_{y_1} \cap \dots \cap U_{y_n}.$$

Then U is open, $x \in U$, and $U \cap V = \emptyset$. □

Corollary 2. *A compact subset of a Hausdorff space is closed.*

Proof. Let $S \subseteq X$ be compact and $x \in X \setminus S$. By the lemma, there exists an open set U containing x with $U \cap S = \emptyset$. Hence $X \setminus S$ is open, so S is closed. □

Theorem 17 (Theorem 6.5). *Every compact Hausdorff space is normal.*

Proof. Let $S, T \subseteq X$ be disjoint closed sets. Since X is compact, S and T are compact. For each $x \in T$, by the lemma on separating a point from a compact set, there exist disjoint open sets U_x, V_x such that

$$x \in U_x, \quad S \subseteq V_x.$$

The family $\{U_x\}_{x \in T}$ is an open cover of T , so by compactness there exist $x_1, \dots, x_n \in T$ such that

$$T \subseteq U_{x_1} \cup \dots \cup U_{x_n} =: U.$$

Set

$$V := V_{x_1} \cap \dots \cap V_{x_n}.$$

Then U and V are open, $T \subseteq U$, $S \subseteq V$, and $U \cap V = \emptyset$. Hence X is normal. □

Theorem 18 (Theorem 6.6). *Let $f : X \rightarrow Y$ be continuous, where X is compact and Y is a topological space. Then $f(X)$ is compact in Y .*

Proof. Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of $f(X)$. Then

$$X = \bigcup_{\alpha \in A} f^{-1}(U_\alpha),$$

and each $f^{-1}(U_\alpha)$ is open in X by continuity of f . Since X is compact, there exist $\alpha_1, \dots, \alpha_n \in A$ such that

$$X = f^{-1}(U_{\alpha_1}) \cup \dots \cup f^{-1}(U_{\alpha_n}).$$

Applying f , we get

$$f(X) \subseteq U_{\alpha_1} \cup \dots \cup U_{\alpha_n}.$$

Thus $\{U_{\alpha_1}, \dots, U_{\alpha_n}\}$ is a finite subcover of $f(X)$, so $f(X)$ is compact. □

Theorem 19 (Theorem 6.7). *Let $f : X \rightarrow Y$ be a continuous map from a compact space X to a Hausdorff space Y . If f is one-to-one, then f is a homeomorphism of X onto $f(X)$.*

Proof. Since f is continuous and bijective onto $f(X)$, it suffices to show that $f^{-1} : f(X) \rightarrow X$ is continuous, or equivalently that $f(U)$ is open in $f(X)$ whenever U is open in X .

Let $U \subseteq X$ be open. Then $X \setminus U$ is closed in X , hence compact. By continuity, $f(X \setminus U)$ is compact in Y , and since Y is Hausdorff, it is closed in Y . Therefore it is closed in $f(X)$.

But

$$f(X \setminus U) = f(X) \setminus f(U)$$

since f is injective. Hence $f(U)$ is open in $f(X)$, and f^{-1} is continuous. Thus f is a homeomorphism onto $f(X)$. $\square$

Arzelà–Ascoli Theorem

Lemma 1. *Let X be compact, and let $(f_n) \subset C(X)$ be uniformly bounded and equicontinuous. Then there exists a subsequence (f_{n_k}) which converges uniformly on X .*

Proof. For each $m \in \mathbb{N}$ and each $x \in X$, equicontinuity gives an open neighbourhood $U_{x,m}$ such that

$$|f_n(y) - f_n(x)| < \frac{1}{m} \quad \text{for all } y \in U_{x,m}, \ n \in \mathbb{N}.$$

By compactness, for each m there exist points $x_{m,1}, \dots, x_{m,r_m}$ such that

$$X = \bigcup_{i=1}^{r_m} U_{x_{m,i},m}.$$

Let

$$A := \bigcup_{m=1}^{\infty} \{x_{m,1}, \dots, x_{m,r_m}\}.$$

Since this is a countable union of finite sets, we may write

$$A = \{a_1, a_2, a_3, \dots\}.$$

Since (f_n) is uniformly bounded, for each j the sequence $(f_n(a_j))$ is bounded, hence admits a convergent subsequence. We now construct nested subsequences

$$(f_n) \supset (f_n^{(1)}) \supset (f_n^{(2)}) \supset (f_n^{(3)}) \supset \dots$$

such that $(f_n^{(k)}(a_k))$ converges for each k .

$$\begin{aligned} (f_n^{(1)}) &= (f_{n_1}^{(1)}, f_{n_2}^{(1)}, f_{n_3}^{(1)}, \dots), \\ (f_n^{(2)}) &= (f_{n_1}^{(2)}, f_{n_2}^{(2)}, f_{n_3}^{(2)}, \dots), \\ (f_n^{(3)}) &= (f_{n_1}^{(3)}, f_{n_2}^{(3)}, f_{n_3}^{(3)}, \dots), \\ &\vdots \end{aligned}$$

where each row is a subsequence of the previous one.

Define the diagonal subsequence by $g_k := f_{n_k}^{(k)}$. Fix j . For all $k \geq j$, the term g_k lies in the subsequence $(f_n^{(j)})$, hence $(g_k(a_j))$ converges. Thus $(g_k(a_j))$ converges for every $a_j \in A$.

We now show that (g_k) is uniformly Cauchy. Let $\varepsilon > 0$ and choose m such that $\frac{3}{m} < \varepsilon$. From the finite cover

$$X = \bigcup_{i=1}^{r_m} U_{x_{m,i},m},$$

and since each $x_{m,i} \in A$, the sequences $(g_k(x_{m,i}))$ converge. Hence there exists N such that for all $k, \ell \geq N$,

$$|g_k(x_{m,i}) - g_\ell(x_{m,i})| < \frac{1}{m} \quad \text{for } i = 1, \dots, r_m.$$

Now let $x \in X$. Choose i such that $x \in U_{x_{m,i},m}$. Then

$$\begin{aligned} |g_k(x) - g_\ell(x)| &\leq |g_k(x) - g_k(x_{m,i})| + |g_k(x_{m,i}) - g_\ell(x_{m,i})| + |g_\ell(x_{m,i}) - g_\ell(x)| \\ &< \frac{1}{m} + \frac{1}{m} + \frac{1}{m} = \frac{3}{m} \\ &< \varepsilon \end{aligned}$$

Since x arbitrary (g_k) is uniformly Cauchy, hence converges uniformly on X . $\square$

Theorem 20 (Arzelà–Ascoli). *Let X be a compact topological space and let $\mathcal{F} \subset C(X)$. Then $\overline{\mathcal{F}}$ is compact in $(C(X), \|\cdot\|_\infty) \Leftrightarrow \mathcal{F}$ is uniformly bounded and equicontinuous.*

Proof. ($\Leftarrow$) Assume $\mathcal{F}$ is uniformly bounded and equicontinuous. Then by the lemma, every sequence in $\mathcal{F}$ has a uniformly convergent subsequence, hence $\overline{\mathcal{F}}$ sequentially compact.

($\Rightarrow$) Assume $\mathcal{F}$ is precompact. Then $\mathcal{F}$ is totally bounded.

Uniform boundedness: For $\varepsilon = 1$, there exist $f_1, \dots, f_n$ such that

$$\mathcal{F} \subset \bigcup_{k=1}^n B(f_k, 1).$$

Hence $\|f\|_\infty \leq 1 + \max_k \|f_k\|_\infty$ for all $f \in \mathcal{F}$.

Equicontinuity: Let $\varepsilon > 0$ and choose $f_1, \dots, f_n$ such that

$$\mathcal{F} \subset \bigcup_{k=1}^n B(f_k, \varepsilon).$$

Fix $x \in X$. Since each f_k is continuous, there exists a neighbourhood U of x such that

$$|f_k(y) - f_k(x)| < \varepsilon \quad \forall y \in U, \quad k = 1, \dots, n.$$

If $f \in \mathcal{F}$, choose k with $\|f - f_k\|_\infty < \varepsilon$. Then

$$|f(y) - f(x)| \leq |f(y) - f_k(y)| + |f_k(y) - f_k(x)| + |f_k(x) - f(x)| < \varepsilon + \varepsilon + \varepsilon = 3\varepsilon.$$

$\square$

Locally Compact

Theorem 21 (One-point compactification). *Let X be a locally compact Hausdorff space. Then there exists a space $Y = X \cup \{\infty\}$ and a topology on Y such that:*

- Y is compact and Hausdorff,
- the subspace topology on X coincides with the original topology,
- the topology is uniquely determined.

The open sets in Y are:

- all open sets of X ,
- all sets of the form $U \cup \{\infty\}$ where $X \setminus U$ is compact in X .

Connectedness

Theorem 22 (Theorem 8.1). *Let f be a continuous function from a connected topological space X to a topological space Y . Then $f(X)$ is connected.*

Proof. Let $E \subseteq f(X)$ be both open and closed in $f(X)$. Then $f^{-1}(E)$ is both open and closed in X since f is continuous.

Because X is connected, we must have either $f^{-1}(E) = \emptyset$ or $f^{-1}(E) = X$.

In the first case, $E = \emptyset$. In the second case, $E = f(X)$. Hence $f(X)$ is connected. $\square$

Theorem 23 (Theorem 8.2). *Let $\{E_\alpha\}_{\alpha \in A}$ be a family of connected subsets of a topological space X such that $E_\alpha \cap E_\beta \neq \emptyset$ for all α, β . Then $\bigcup_{\alpha \in A} E_\alpha$ is connected.*

Proof. Let $E = \bigcup_{\alpha} E_\alpha$ and let $F \subseteq E$ be clopen with $F \neq \emptyset$. Pick $x \in F \cap E_{\alpha_0}$. Since E_{α_0} is connected, $F \cap E_{\alpha_0} = E_{\alpha_0}$, so $E_{\alpha_0} \subseteq F$.

For any β , $F \cap E_\beta$ is clopen in E_β and contains $E_{\alpha_0} \cap E_\beta \neq \emptyset$, hence $E_\beta \subseteq F$. Thus $F = E$. $\square$

Definition 16. *Let X be a topological space and $x \in X$. The connected component of x , denoted $C(x)$, is the union of all connected subsets of X that contain x .*

Theorem 24. *$C(x)$ is connected and is the largest connected subset of X containing x .*

Theorem 25 (Theorem 8.3). *Two connected components of X either coincide or are disjoint. Hence, the connected components form a partition of X .*

Theorem 26 (Theorem 8.4). *Any interval in $\mathbb{R}$ (closed, open, or half-open; finite or infinite) is connected.*

Proof. It suffices to show that $[0, 1]$ is connected. Let $E \subseteq [0, 1]$ be clopen and assume $E \neq \emptyset$. Set

$$t = \sup E.$$

Since E is closed, $t \in E$. If $t < 1$, then since E is open, there exists $\varepsilon > 0$ such that $(t, t + \varepsilon) \subseteq E$, contradicting the definition of t . Hence $E = [0, 1]$. $\square$

Path Connectedness

Theorem 27 (Lemma 9.1). *The relation “there is a path in X from x to y ” is an equivalence relation on X .*

Proof. We verify the three properties:

(i) **Reflexive:** For any $x \in X$, the constant map

$$\alpha : [0, 1] \rightarrow X, \quad \alpha(s) = x$$

is a path from x to x .

(ii) **Symmetric:** If α is a path from x to y , then

$$\beta : [0, 1] \rightarrow X, \quad \beta(s) = \alpha(1 - s)$$

is a path from y to x .

(iii) **Transitive:** If α is a path from x to y and β is a path from y to z , define

$$\gamma(s) = \begin{cases} \alpha(2s), & 0 \leq s \leq \frac{1}{2}, \\ \beta(2s - 1), & \frac{1}{2} \leq s \leq 1. \end{cases}$$

Then γ is a path from x to z .

Hence the relation is an equivalence relation. □

Theorem 28 (Theorem 9.2). *Every path-connected space is connected.*

Proof. Fix $x_0 \in X$. For each $x \in X$, let $\gamma_x : [0, 1] \rightarrow X$ be a path from x_0 to x . Since $[0, 1]$ is connected and γ_x is continuous, $\gamma_x([0, 1])$ is connected. Also, each $\gamma_x([0, 1])$ contains x_0 . Hence

$$X = \bigcup_{x \in X} \gamma_x([0, 1])$$

is a union of connected subsets with a common point, so X is connected. □

Corollary 3 (Corollary 9.3). *Each connected component of a topological space is a union of path components.*

Proof. A path component is path-connected, hence connected by the theorem. Therefore each path component is contained in some connected component. It follows that a connected component is the union of all path components contained in it. □

Finite Product Spaces

Theorem 29 (10.1). *Let X be the product of the topological spaces $X_1, \dots, X_n$, and let π_j be the projection of X onto X_j . The product topology for X is the smallest topology for which each of the projections π_j is continuous.*

Proof. Let $\mathcal{T}$ be another topology for X such that the projections π_j are $\mathcal{T}$ -continuous. Let $U_1, \dots, U_n$ be open subsets of $X_1, \dots, X_n$, respectively. Then each $\pi_j^{-1}(U_j)$ belongs to $\mathcal{T}$. Since

$$\pi_1^{-1}(U_1) \cap \dots \cap \pi_n^{-1}(U_n) = U_1 \times \dots \times U_n,$$

the basic set $U_1 \times \dots \times U_n$ belongs to $\mathcal{T}$ and $\mathcal{T}$ includes the product topology. $\square$

Theorem 30 (10.2). *Let $X = X_1 \times \dots \times X_n$ be the product of the topological spaces $X_1, \dots, X_n$. Then each projection π_j is an open map of X onto X_j .*

Proof. If $U = U_1 \times \dots \times U_n$ is a nonempty basic open set, then $\pi_j(U) = U_j$ is open. Since maps preserve unions, the image of any open set is open. $\square$

Theorem 31 (10.3). *Let E be a topological space and let f be a function from E to the product*

$$X = X_1 \times \dots \times X_n.$$

Then f is continuous if and only if each $\pi_j \circ f$ is continuous, $1 \leq j \leq n$.

Proof. If f is continuous, then $\pi_j \circ f$ is also continuous since the composition of continuous functions is continuous.

Conversely, suppose that each $\pi_j \circ f$ is continuous. Let

$$U = U_1 \times \dots \times U_n$$

be a basic open set in X . Then

$$f^{-1}(U) = (\pi_1 \circ f)^{-1}(U_1) \cap \dots \cap (\pi_n \circ f)^{-1}(U_n)$$

is a finite intersection of open sets and hence is open. Since inverses of functions preserve unions, the inverse image of any open set is open. Hence f is continuous. $\square$

Theorem 32 (10.4). *If $X_1, \dots, X_n$ are Hausdorff spaces, then $X = X_1 \times \dots \times X_n$ is Hausdorff.*

Proof. Let $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ be distinct points of X . Choose an index j such that $x_j \neq y_j$. Let U_j and V_j be disjoint open neighborhoods of x_j and y_j , respectively, in X_j . Then $\pi_j^{-1}(U_j)$ and $\pi_j^{-1}(V_j)$ are disjoint open neighborhoods of x and y , respectively, in X . $\square$

Theorem 33 (10.5). *If X_j is a path-connected topological space, $1 \leq j \leq n$, then $X = X_1 \times \dots \times X_n$ is path-connected.*

Proof. Let $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ be points of X . For $1 \leq j \leq n$, let

$$\gamma_j : [0, 1] \rightarrow X_j$$

be a path from x_j to y_j . Then the path

$$\gamma : [0, 1] \rightarrow X, \quad \gamma(t) = (\gamma_1(t), \dots, \gamma_n(t)), \quad 0 \leq t \leq 1,$$

is a path from x to y . Hence X is path-connected. $\square$

Theorem 34 (10.6). *If X_j is a connected topological space, $1 \leq j \leq n$, then $X = X_1 \times \dots \times X_n$ is also connected.*

Proof. Let E be a closed and open subset of X . Replacing E by its complement, if necessary, we can assume that $E \neq \emptyset$. We must show that $E = X$.

Let $z = (z_1, \dots, z_n) \in E$ and let $y = (y_1, \dots, y_n) \in X$. The intersection

$$E \cap (X_1 \times \{z_2\} \times \dots \times \{z_n\})$$

is a relatively open and closed subset of the coordinate slice, which includes z . Since X_1 is connected, so is the coordinate slice. It follows that E must include the entire coordinate slice. In particular,

$$(y_1, z_2, \dots, z_n) \in E.$$

Now we repeat this argument in the second coordinate and continue until eventually we conclude that $y \in E$. Since $y \in X$ is arbitrary, we obtain $E = X$. $\square$

Lemma 2 (10.7). *Let Y be a topological space and let $\mathcal{B}$ be a base for the topology of Y . If every open cover of Y by sets in $\mathcal{B}$ has a finite subcover, then Y is compact.*

Proof. Let $\{U_\beta\}_{\beta \in B}$ be an open cover of Y . For each $y \in Y$, choose $V_y \in \mathcal{B}$ and an index β such that

$$y \in V_y \subset U_\beta.$$

The family $\{V_y : y \in Y\}$ forms an open cover of Y by sets in $\mathcal{B}$. By hypothesis, there exist a finite number of the V_y 's that cover Y . Since each of these V_y 's is contained in at least one of the U_β 's, we obtain a finite number of the U_β 's that cover Y . Hence Y is compact. $\square$

Theorem 35 (10.8, Tychonoff's Theorem, finite case). *If $X_1, \dots, X_n$ are compact spaces, then $X_1 \times \dots \times X_n$ is compact.*

Proof. It suffices to prove the theorem in the case where $n = 2$, and so we consider only

$$X = X_1 \times X_2.$$

Let $\mathcal{C}$ be a cover of $X_1 \times X_2$ by rectangular open sets of the form $U \times V$, where U is open in X_1 and V is open in X_2 . By Lemma 10.7, it suffices to show that the cover $\mathcal{C}$ has a finite subcover.

Fix $z \in X_2$. The coordinate slice $X_1 \times \{z\}$ is compact. Hence there are a finite number of sets

$$U_1 \times V_1, \dots, U_m \times V_m \in \mathcal{C}$$

that cover $X_1 \times \{z\}$. We can assume that $z \in V_j$ for all j , $1 \leq j \leq m$. Then

$$V(z) = V_1 \cap \dots \cap V_m$$

is an open neighborhood of z in X_2 . Furthermore,

$$\pi_2^{-1}(V(z))$$

is covered by a finite number of sets in $\mathcal{C}$, namely the sets

$$U_j \times V_j, \quad 1 \leq j \leq m.$$

The sets $\{V(z) : z \in X_2\}$ form an open cover of X_2 . Since X_2 is compact, there exist

$$z_1, \dots, z_k \in X_2$$

such that

$$X_2 = V(z_1) \cup \cdots \cup V(z_k).$$

Then

$$X = \pi_2^{-1}(V(z_1)) \cup \cdots \cup \pi_2^{-1}(V(z_k)).$$

Since each $\pi_2^{-1}(V(z_j))$ can be covered by a finite number of sets in $\mathcal{C}$, and since the aggregate of these sets covers X , $\mathcal{C}$ has a finite subcover. $\square$

Axiom of Choice & Zorn's Lemma

Definition 17 (Axiom of Choice). *Let $\{U_\alpha\}_{\alpha \in A} \subset \mathcal{P}(X)$ be a family of non-empty sets indexed by some set A . Then there exists a function $f : A \rightarrow X$ such that $f(\alpha) \in U_\alpha$ for all $\alpha \in A$.*

Definition 18 (Partially Ordered Set). *A set P together with a relation $\leq$ is called a partially ordered set if for all $x, y, z \in P$:*

- $x \leq x$ (reflexivity),
- if $x \leq y$ and $y \leq x$, then $x = y$ (antisymmetry),
- if $x \leq y$ and $y \leq z$, then $x \leq z$ (transitivity).

Definition 19 (Totally Ordered Subset). *A subset $C \subset P$ is called totally ordered (or a chain) if for all $x, y \in C$, either $x \leq y$ or $y \leq x$.*

Definition 20 (Upper Bound). *Let $C \subset P$. An element $u \in P$ is called an upper bound of C if*

$$x \leq u \quad \text{for all } x \in C.$$

Definition 21 (Maximal Element). *An element $m \in P$ is called maximal if there is no $x \in P$ such that*

$$m < x.$$

Lemma 3 (Zorn's Lemma). *Let P be a partially ordered set. If each totally ordered subset of P has an upper bound, then P has a maximal element.*

Remark 3 (Interpretation and Applications of Zorn's Lemma). *Zorn's Lemma is a fundamental existence principle. It allows one to construct maximal objects in situations where a direct construction is not possible.*

The typical strategy is:

1. Define a partially ordered set of "partial objects" (e.g. partial functions, partial covers, or substructures),
2. Show that every chain has an upper bound (i.e. partial objects can be extended),
3. Conclude the existence of a maximal element.

In this course, Zorn's Lemma appears in the proof of Tychonoff's Theorem, where it is used to construct a maximal open cover without a finite subcover, leading to a contradiction.

More generally, Zorn's Lemma is equivalent to the Axiom of Choice and is widely used throughout mathematics, for example in proving the existence of bases of vector spaces, maximal ideals, and extensions of partial functions.

Infinite Product Spaces

Theorem 36 (12.3, Alexander Subbase Theorem). *Let Y be a topological space and let $\mathcal{S}$ be a subbase for the topology of Y . If every open cover of Y by sets in $\mathcal{S}$ has a finite subcover, then Y is compact.*

Proof. Let $\mathcal{C}$ be an open cover of Y such that no finite subfamily of $\mathcal{C}$ covers Y . It suffices to show that there exists a cover of Y by sets in $\mathcal{S}$ that has no finite subcover.

Consider families $\mathcal{B}$ of open subsets of Y such that $\mathcal{C} \subset \mathcal{B}$ and such that no finite collection of sets in $\mathcal{B}$ covers Y . The aggregate of such families forms a partially ordered set, with the ordering of inclusion. Let $\{\mathcal{B}_\beta\}_{\beta \in B}$ be a totally ordered subset of $\mathcal{B}$. Let $\mathcal{B}_0$ be the family of subsets of Y that is the union of the $\mathcal{B}_\beta$'s. We claim that $\mathcal{B}_0 \in \mathcal{B}$, that is, that no finite collection of sets in $\mathcal{B}_0$ covers Y . Indeed, any finite collection of sets in $\mathcal{B}_0$ is included in one of the $\mathcal{B}_\beta$'s, because of the total ordering, and no finite family of sets in any $\mathcal{B}_\beta$ covers Y . (Compare this argument with the corresponding proof that every cover has a basis.) Evidently $\mathcal{B}_0$ is an upper bound for $\{\mathcal{B}_\beta\}$. Hence the hypothesis of Zorn's Lemma are met. By Zorn's Lemma, there is a maximal element $\mathcal{B}_0$ in $\mathcal{B}$.

It suffices now to show that the family $\mathcal{S}$ covers Y . Indeed, by the defining property of $\mathcal{B}_0$, no finite subfamily of $\mathcal{B}_0$ covers Y .

Let $x \in Y$. It suffices to show that there exists $V \in \mathcal{S}$ such that $x \in V$. Choose $U \in \mathcal{B}_0$ such that $x \in U$. Since $\mathcal{S}$ is a subbase, there exist $V_1, \dots, V_n \in \mathcal{S}$ such that

$$x \in V_1 \cap \dots \cap V_n \subset U.$$

Suppose that no V_j belongs to $\mathcal{B}_0$. Because of the maximality of $\mathcal{B}_0$, there exist for each j a finite number of open sets

$$W_{j1}, \dots, W_{jk_j} \in \mathcal{B}_0$$

such that

$$W_{j1} \cup \dots \cup W_{jk_j} \cup V_j$$

covers Y . Otherwise we could add V_j to $\mathcal{B}_0$, contradicting the maximality of $\mathcal{B}_0$. Then

$$(V_1 \cap \dots \cap V_n) \cup \bigcup_{j=1}^n \bigcup_{i=1}^{k_j} W_{ji}$$

covers Y . Hence

$$U \cup \bigcup_{j=1}^n \bigcup_{i=1}^{k_j} W_{ji}$$

covers Y . This gives a finite cover of Y by sets in $\mathcal{B}_0$, contradicting the fact that $\mathcal{B}_0$ belongs to $\mathcal{B}$. We conclude that some V_j belongs to $\mathcal{B}_0$, and we take this V_j to cover x . $\square$

Theorem 37 (12.4, Tychonoff's Theorem). *Any product of compact spaces is compact.*

Proof. Let

$$X = \prod_{\alpha \in A} X_\alpha,$$

where each X_α is compact. Let $\mathcal{S}$ be a collection of subbasic open sets in the product topology on X such that no finite subcollection of $\mathcal{S}$ covers X . By Alexander's Subbase Theorem, it suffices to show that $\mathcal{S}$ does not cover X .

Each element of $\mathcal{S}$ is of the form

$$\pi_\alpha^{-1}(U),$$

for some $\alpha \in A$ and some open set $U \subseteq X_\alpha$. Now fix $\beta \in A$, and define

$$\mathcal{U}_\beta = \{U \subseteq X_\beta : U \text{ is open in } X_\beta \text{ and } \pi_\beta^{-1}(U) \in \mathcal{S}\}.$$

We claim that $\mathcal{U}_\beta$ does not cover X_β . For otherwise, $\mathcal{U}_\beta$ would be an open cover of X_β , and since X_β is compact, there would exist $U_1, \dots, U_n \in \mathcal{U}_\beta$ such that

$$X_\beta = U_1 \cup \dots \cup U_n.$$

But then

$$X = \pi_\beta^{-1}(U_1) \cup \dots \cup \pi_\beta^{-1}(U_n),$$

which is a finite cover of X by members of $\mathcal{S}$, contradicting the choice of $\mathcal{S}$. Hence $\mathcal{U}_\beta$ does not cover X_β , so we may choose

$$x_\beta \in X_\beta \setminus \bigcup_{U \in \mathcal{U}_\beta} U.$$

Doing this for each $\beta \in A$, we obtain a point

$$x = (x_\alpha)_{\alpha \in A} \in \prod_{\alpha \in A} X_\alpha.$$

We show that $x \notin S$ for every $S \in \mathcal{S}$. Indeed, if

$$S = \pi_\alpha^{-1}(U)$$

for some $\alpha \in A$ and open $U \subseteq X_\alpha$, then $U \in \mathcal{U}_\alpha$, so $x_\alpha \notin U$ by construction. Therefore

$$x \notin \pi_\alpha^{-1}(U) = S.$$

Thus x is not contained in any member of $\mathcal{S}$, and so $\mathcal{S}$ does not cover X . □

Stone–Weierstrass Approximation Theorem

Definition 22 (Lattice operations). *For functions $f, g \in C(X, \mathbb{R})$, define the join and meet by*

$$(f \vee g)(x) = \max\{f(x), g(x)\}, \quad (f \wedge g)(x) = \min\{f(x), g(x)\}.$$

A subset $L \subset C(X, \mathbb{R})$ is called a lattice if for all $f, g \in L$,

$$f \vee g \in L \quad \text{and} \quad f \wedge g \in L.$$

If L is also a subspace (or subalgebra) of $C(X, \mathbb{R})$, then L is called a sublattice.

Lemma 4. *Let X be a compact Hausdorff space with more than one point, and let L be a closed sublattice of $C(X, \mathbb{R})$ with the following property: for any distinct $x, y \in X$ and any $a, b \in \mathbb{R}$, there exists $h \in L$ such that $h(x) = a, h(y) = b$. Then $L = C(X, \mathbb{R})$.*

Proof. Let $f \in C(X, \mathbb{R})$ and let $\varepsilon > 0$. Since L is closed in $C(X, \mathbb{R})$, it is enough to find $g \in L$ such that

$$f(z) - \varepsilon < g(z) < f(z) + \varepsilon \quad \text{for all } z \in X.$$

Fix $x \in X$. For each $y \neq x$, by hypothesis there exists $f_y \in L$ such that

$$f_y(x) = f(x), \quad f_y(y) = f(y).$$

Consider the open set

$$G_y = \{z \in X : f_y(z) < f(z) + \varepsilon\}.$$

Since $x, y \in G_y$, the family $\{G_y : y \neq x\}$ is an open cover of X . By compactness, choose $y_1, \dots, y_n$ such that

$$X = G_{y_1} \cup \dots \cup G_{y_n}.$$

Set

$$g_x = f_{y_1} \wedge \dots \wedge f_{y_n}.$$

Because L is a sublattice, $g_x \in L$. Also,

$$g_x(x) = f(x),$$

and for every $z \in X$ there exists some j with $z \in G_{y_j}$, hence

$$g_x(z) \leq f_{y_j}(z) < f(z) + \varepsilon.$$

Now define

$$H_x = \{z \in X : g_x(z) > f(z) - \varepsilon\}.$$

Since $g_x(x) = f(x)$, we have $x \in H_x$, so the sets $\{H_x : x \in X\}$ form an open cover of X . By compactness, choose $x_1, \dots, x_m$ such that

$$X = H_{x_1} \cup \dots \cup H_{x_m}.$$

Finally set

$$g = g_{x_1} \vee \dots \vee g_{x_m}.$$

Again $g \in L$. For any $z \in X$, since $z \in H_{x_i}$ for some i ,

$$g(z) \geq g_{x_i}(z) > f(z) - \varepsilon,$$

while for every i we already know $g_{x_i}(z) < f(z) + \varepsilon$, so

$$g(z) < f(z) + \varepsilon.$$

Thus

$$f(z) - \varepsilon < g(z) < f(z) + \varepsilon \quad \text{for all } z \in X.$$

Hence $\|f - g\|_\infty < \varepsilon$. Since L is closed and f can be uniformly approximated by elements of L , it follows that $f \in L$. Therefore $L = C(X, \mathbb{R})$. $\square$

Lemma 5. *Let X be a topological space. Then every closed subalgebra $A \subset C(X, \mathbb{R})$ is also a closed sublattice of $C(X, \mathbb{R})$.*

Proof. It suffices to show that if $f \in A$, then $|f| \in A$. Indeed, for $f, g \in A$, the lattice operations can be written as

$$f \vee g = \frac{f + g + |f - g|}{2}, \quad f \wedge g = \frac{f + g - |f - g|}{2}.$$

Since A is a subalgebra, it is closed under addition and scalar multiplication, so if $|f - g| \in A$, then $f \vee g, f \wedge g \in A$. Thus it remains to prove that $|f| \in A$ for every $f \in A$.

Let $f \in A$ and $\varepsilon > 0$. Since f is continuous and X is arbitrary, the function $|t|$ on the interval $[-\|f\|_\infty, \|f\|_\infty]$ can be uniformly approximated by polynomials. Hence there exists a polynomial p such that

$$|t| - p(t) < \varepsilon \quad \text{for all } t \in [-\|f\|_\infty, \|f\|_\infty].$$

By adjusting the constant term, we may assume $p(0) = 0$.

Since A is an algebra, $p(f) \in A$. Moreover, for all $x \in X$,

$$||f(x)| - p(f(x))| < \varepsilon,$$

so $\||f| - p(f)\|_\infty < \varepsilon$. Since A is closed in $C(X, \mathbb{R})$, it follows that $|f| \in A$.

Therefore A is closed under $|\cdot|$, and hence under $\vee$ and $\wedge$, so A is a sublattice. $\square$

Theorem 38 (Real Stone–Weierstrass Theorem). *Let X be a compact Hausdorff space, and let A be a closed subalgebra of $C(X, \mathbb{R})$ which separates points and contains a non-zero constant function. Then $A = C(X, \mathbb{R})$.*

Proof. If X has only one point, then $C(X, \mathbb{R})$ contains only constant functions; and since A contains a non-zero constant function and is an algebra, it contains all constant functions and equals $C(X, \mathbb{R})$. We may thus assume that X has more than one point.

By the preceding lemmas, it suffices to show that if x and y are distinct points of X , and if a and b are any two real numbers, then there exists a function $f \in A$ such that

$$f(x) = a \quad \text{and} \quad f(y) = b.$$

Since A separates points, there exists a function $g \in A$ such that $g(x) \neq g(y)$. Define

$$f(z) = a \frac{g(z) - g(y)}{g(x) - g(y)} + b \frac{g(x) - g(z)}{g(x) - g(y)}.$$

Then $f \in A$ and clearly satisfies $f(x) = a$ and $f(y) = b$. Hence $A = C(X, \mathbb{R})$. $\square$

Theorem 39 (Complex Stone–Weierstrass Theorem). *Let X be a compact Hausdorff space, and let A be a closed subalgebra of $C(X, \mathbb{C})$ which separates points, contains a non-zero constant function, and is closed under complex conjugation, i.e. $f \in A \implies \bar{f} \in A$. Then $A = C(X, \mathbb{C})$.*

Proof. Let

$$B := \{f \in A : f(X) \subset \mathbb{R}\}$$

be the set of real-valued functions in A . We claim that B is a closed subalgebra of $C(X, \mathbb{R})$.

First, if $f, g \in B$ and $\alpha \in \mathbb{R}$, then $f + g$, αf , and fg are again real-valued and belong to A since A is a subalgebra. Hence B is a subalgebra of $C(X, \mathbb{R})$.

Also, B is closed in $C(X, \mathbb{R})$: if (f_n) is a sequence in B converging uniformly to $f \in C(X, \mathbb{R})$, then $f_n \rightarrow f$ also in $C(X, \mathbb{C})$. Since A is closed in $C(X, \mathbb{C})$ and each $f_n \in A$, it follows that $f \in A$. As f is real-valued, we get $f \in B$.

Next we show that B separates points. Let $x, y \in X$ with $x \neq y$. Since A separates points, there exists $f \in A$ such that

$$f(x) \neq f(y).$$

Write

$$f = u + iv,$$

where

$$u = \Re f = \frac{f + \bar{f}}{2}, \quad v = \Im f = \frac{f - \bar{f}}{2i}.$$

Because A is closed under conjugation and is a subalgebra, we have $u, v \in A$. Since u and v are real-valued, actually $u, v \in B$.

Now if both $u(x) = u(y)$ and $v(x) = v(y)$ held, then

$$f(x) = u(x) + iv(x) = u(y) + iv(y) = f(y),$$

a contradiction. Hence either $u(x) \neq u(y)$ or $v(x) \neq v(y)$, so B separates points.

Since A contains a non-zero constant function, say $c \neq 0$, and is closed under conjugation, it also contains $\bar{c}$. Therefore

$$\frac{c + \bar{c}}{2} = \Re(c) \quad \text{and} \quad \frac{c - \bar{c}}{2i} = \Im(c)$$

belong to B , and at least one of them is non-zero. Thus B contains a non-zero real constant function.

Therefore B is a closed subalgebra of $C(X, \mathbb{R})$ which separates points and contains a non-zero constant function. By the real Stone–Weierstrass theorem,

$$B = C(X, \mathbb{R}).$$

It remains to show that $A = C(X, \mathbb{C})$. Let $f \in C(X, \mathbb{C})$. Write

$$f = u + iv,$$

where $u, v \in C(X, \mathbb{R})$. Since $B = C(X, \mathbb{R})$, we have $u, v \in B \subset A$. Because A is closed under multiplication by complex scalars, $iv \in A$, and hence $f = u + iv \in A$. Thus every $f \in C(X, \mathbb{C})$ belongs to $A \implies A = C(X, \mathbb{C})$. $\square$